Chains and Agents

No agents (zero–shot)

Please create an essay on „The Calling of St. Matthew" in the context of the Baroque period

start

finish

No agents (Chain-of-Thought)

start

finish

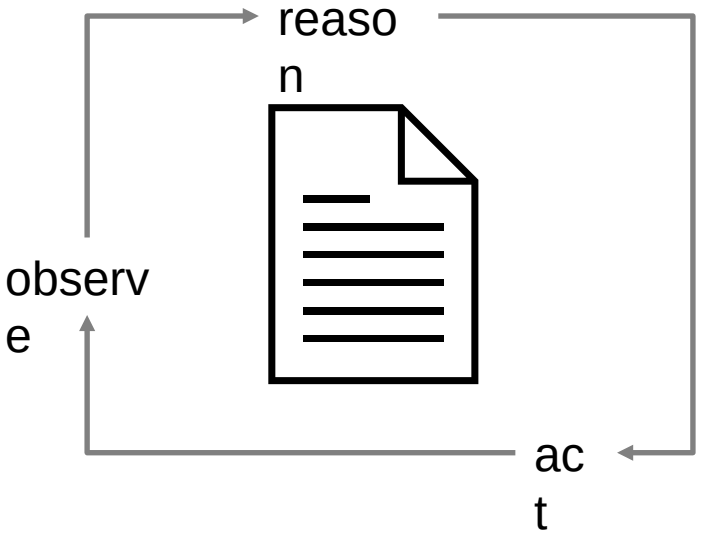
Please create an essay on „The Calling of St. Matthew" in the context of the Baroque period.

Use chain-of-thought reasoning to break down your explanation into logical steps.

Chain-of-Thought Explanation:

Step 1: Identify the Context of the Baroque Period
Step 2: Analyze the Subject Matter of the Painting
Step 3: Explore the Symbolism and Iconography
Step 4: Consider the Impact on Art and Culture

Agentic Workflow



Please create an essay on „The Calling of St. Matthew" in the context of the Baroque period.

Do you need any web research?

Create a first draft

Check it carefully, and identify parts that need reveision or more research

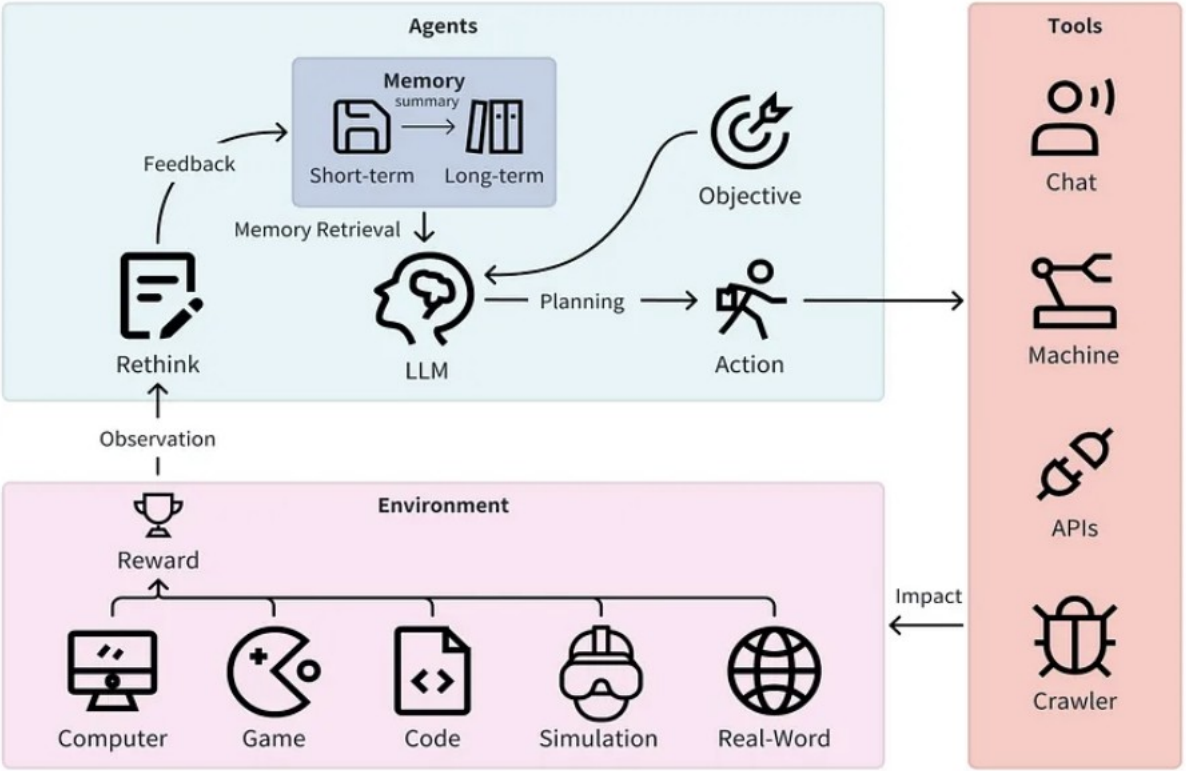
Revise Your draft

Reasoning Design Patterns

1. Reflection
2. Tools
3. Planning
4. Multi-agent

Agents

Reinforcement Loop



https://arxiv.org/pdf/2401.03428

Reflection

Re-Act

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports',' Front Row (software)', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys]

https://arxiv.org/abs/2210.03629

Reflection

Re-Woo: Reasoning WithOut Observation

ReWOO: Decoupling Reasoning from Observations for Efficient Augmented Language Models

Binfeng Xu, Zhiyuan Peng, Bowen Lei, Subhabrata Mukherjee, Yuchen Liu, Dongkuan Xu

Augmented Language Models (ALMs) blend the reasoning capabilities of Large Language Models (LLMs) with tools that allow for knowledge retrieval and action execution. Existing ALM systems trigger LLM thought processes while pulling observations from these tools in an interleaved fashion. Specifically, an LLM reasons to call an external tool, gets halted to fetch the tool's response, and then decides the next action based on all preceding response tokens. Such a paradigm, though straightforward and easy to implement, often leads to huge computation complexity from redundant prompts and repeated execution. This study addresses such challenges for the first time, proposing a modular paradigm ReWOO (Reasoning WithOut Observation) that detaches the reasoning process from external observations, thus significantly reducing token consumption. Comprehensive evaluations across six public NLP benchmarks and a curated dataset reveal consistent performance enhancements with our proposed methodology. Notably, ReWOO achieves 5x token efficiency and 4% accuracy improvement on HotpotQA, a multi-step reasoning benchmark. Furthermore, ReWOO demonstrates robustness under tool-failure scenarios. Beyond prompt efficiency, decoupling parametric modules from non-parametric tool calls enables instruction fine-tuning to offload LLMs into smaller language models, thus substantially reducing model parameters. Our illustrative work offloads reasoning ability from 175B GPT3.5 into 7B LLaMA, demonstrating the significant potential for truly efficient and scalable ALM systems.

https://arxiv.org/abs/2305.18323

Reflection

Re-Woo: Reasoning WithOut Observation

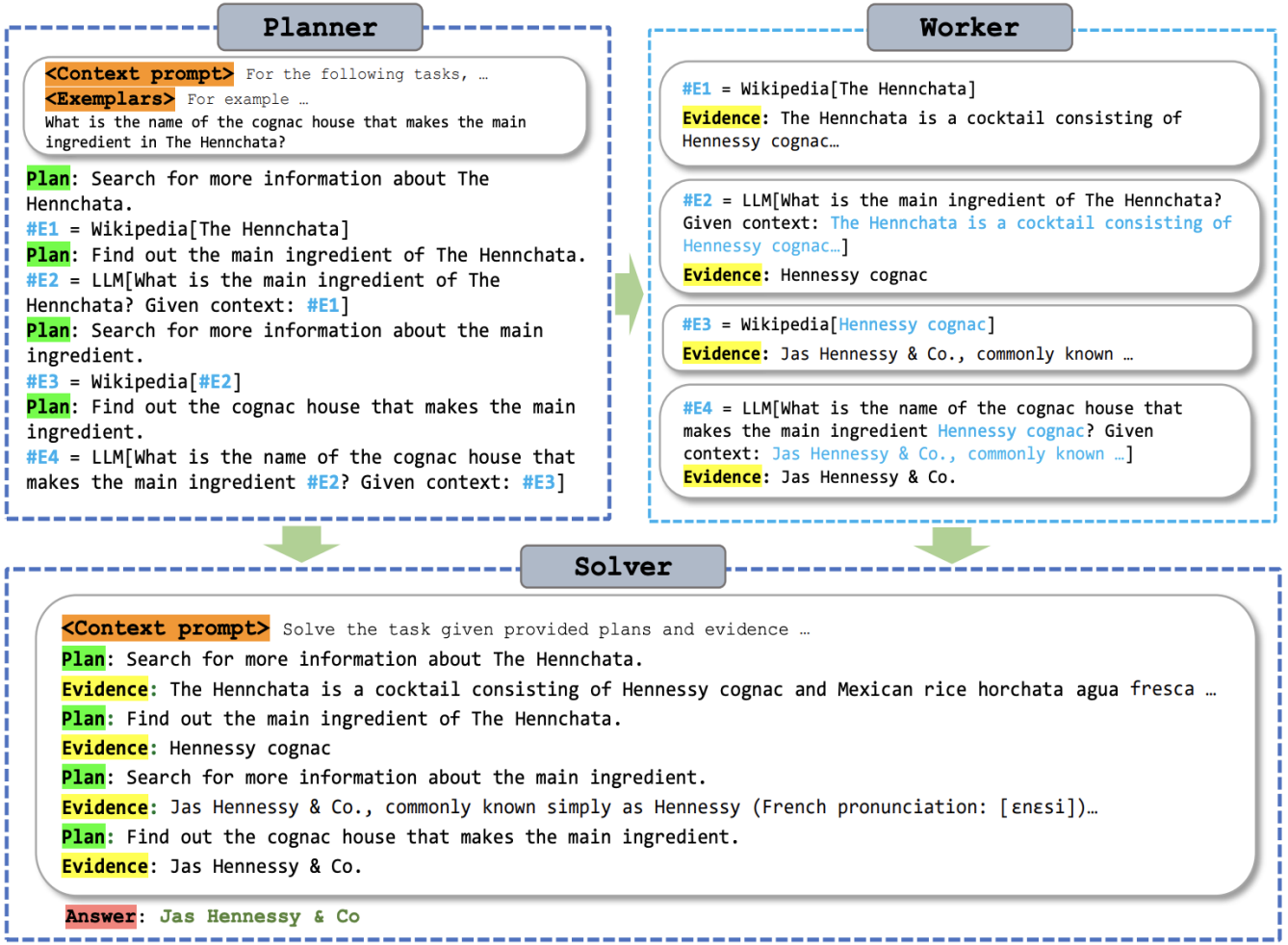
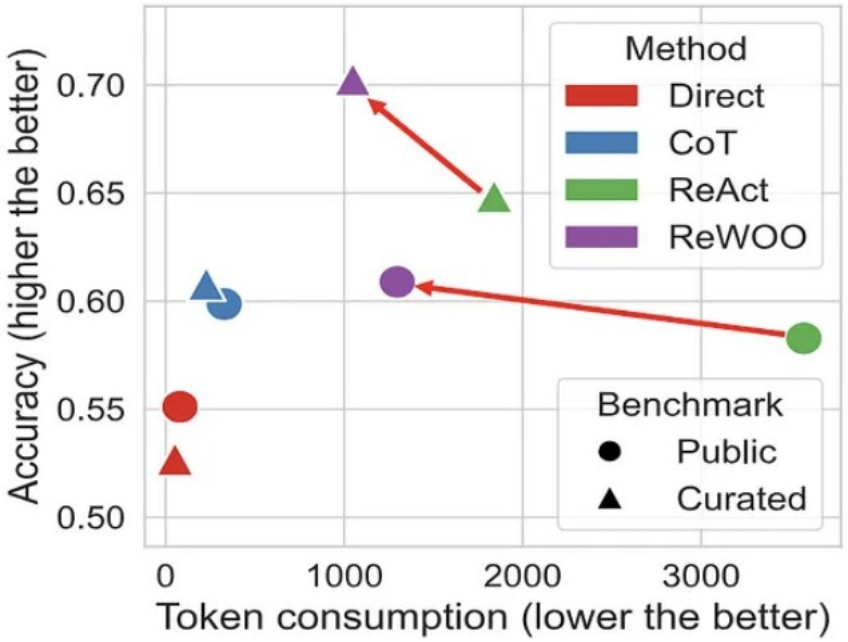


Figure 1: Workflow of ReWOO . Given a question, Planner composes a comprehensive blueprint of interlinked plans prior to tool response. The blueprint instructs Worker to use external tools and collect evidence. Finally, plans and evidence are paired and fed to Solver for the answer.

https://arxiv.org/abs/2305.18323

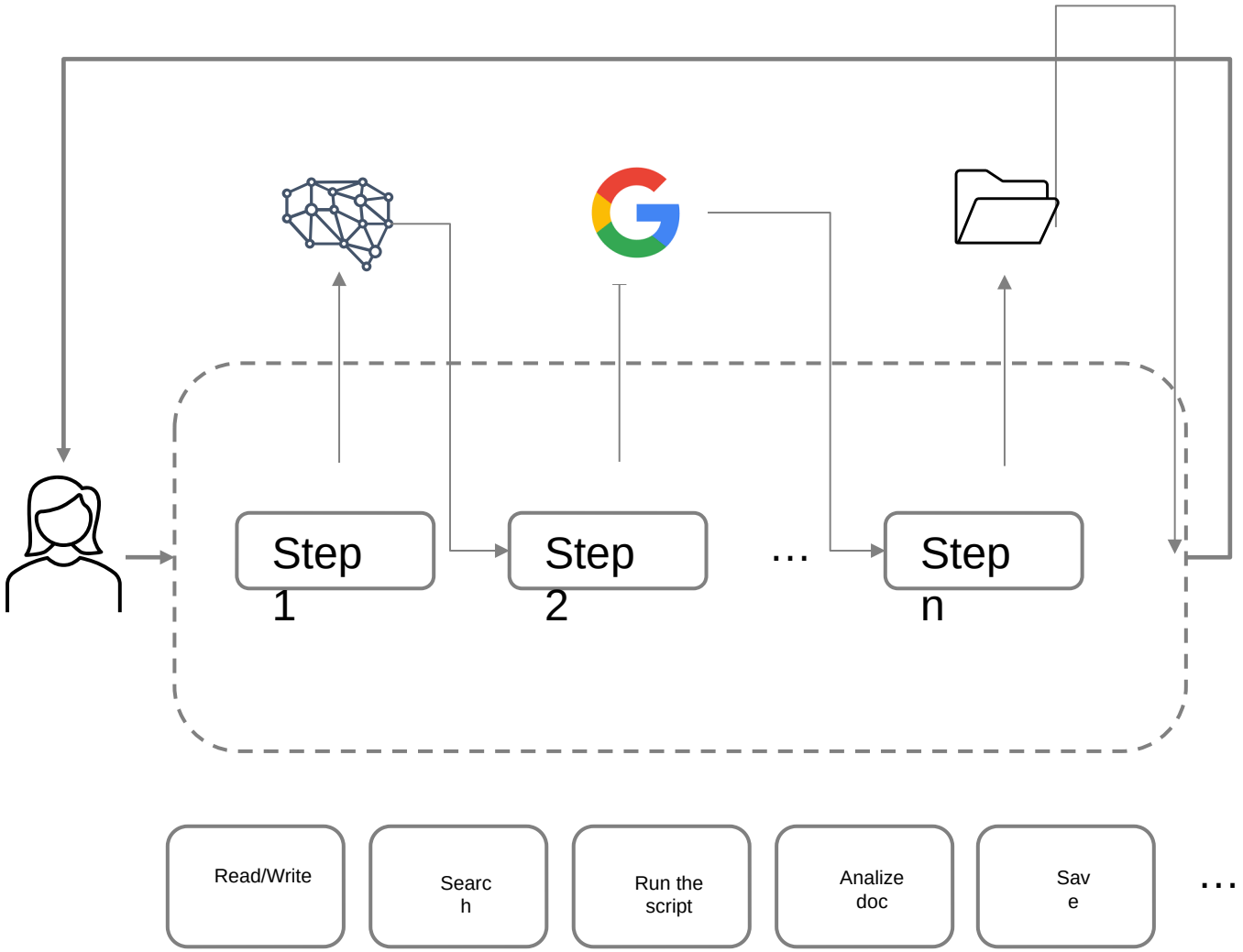
Reflection

Re-Woo: Reasoning WithOut Observation



https://arxiv.org/abs/2305.18323

Tools




```
chain = prompt | step_1 | step_2 | step_n |
output_parser
```

Chains

```python
from langchain_openai import ChatOpenAI
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import RunnablePassthrough


prompt = ChatPromptTemplate.from_template(
    "Tell me a short joke about {topic}"
)
output_parser = StrOutputParser()
model = ChatOpenAI(model="gpt-3.5-turbo")
chain = (
    {"topic": RunnablePassthrough()}
    | prompt
    | model
    | output_parser
)

chain.invoke("ice cream")
```

LCEL

LangChain
Expression
Language

Tools

python.langchain.com/docs/integrations/toolkits/

LangChain Docs Use cases Integrations Guides API More

Providers
- Anthropic
- AWS
- Google
- Hugging Face
- Microsoft
- OpenAI
- More

Components
- LLMs
- Chat models
- Embedding models
- Document loaders
- Document transformers
- Vector stores
- Retrievers
- Tools
- Toolkits
  - AINetwork
  - Airbyte Question Answering
  - Amadeus
  - Azure Cognitive Services
  - ClickUp
  - Cogniswitch Tools
  - Connery Toolkit
  - CSV
  - Document Comparison
  - Github
  - Gitlab
  - Gmail
  - Jira
  - JSON
  - MultiOn
  - NASA
  - Office365
  - OpenAPI
  - Natural Language APIs
  - Pandas Dataframe
  - PlayWright Browser
  - Polygon IO Toolkit
  - PowerBI Dataset
  - Python
  - Robocorp
  - Slack
  - Spark Dataframe
  - Spark SQL
  - SQL Database

Components > Toolkits

Toolkits

AINetwork
AI Network is a layer 1

Airbyte Question Answering
This notebook shows how to do question answering over structur...

Amadeus
This notebook walks you through connecting LangChain to the A...

Azure Cognitive Services
This toolkit is used to interact with the Azure Cognitive Services ...

ClickUp
ClickUp is an all-in-one productivity platform

Cogniswitch Tools
Use CogniSwitch to build production ready applications that can

Connery Toolkit
Using this toolkit, you can integrate Connery Actions into your

CSV
This notebook shows how to use agents to interact with data in C...

Document Comparison
This notebook shows how to use an agent to compare two docum...

Github
The Github toolkit contains tools that enable an LLM agent to inte...

Gitlab
The Gitlab toolkit contains tools that enable an LLM agent to inter...

Gmail
This notebook walks through connecting a LangChain email to the

Jira
This notebook goes over how to use the Jira toolkit.

JSON
This notebook showcases an agent interacting with large JSON/d...

MultiOn
MultiON

NASA
This notebook shows how to use agents to interact with the NAS...

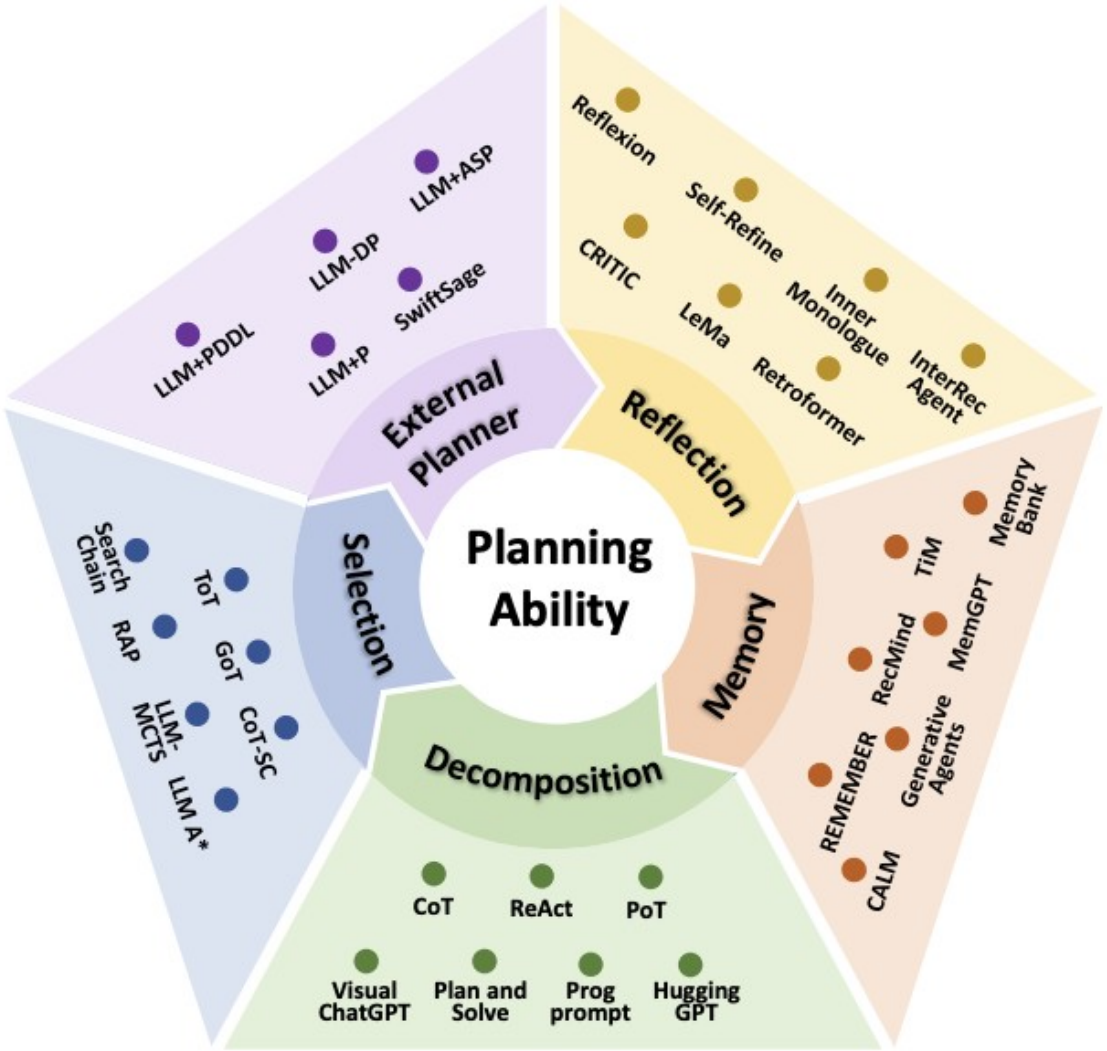
Office365
Microsoft 365 is a product family of

OpenAPI
We can construct agents to consume arbitrary APIs, here APIs co...

Natural Language APIs
Natural Language API Toolkits (NLAToolkits) permit LangChain Ag...

Pandas Dataframe
This notebook shows how to use agents to interact with a

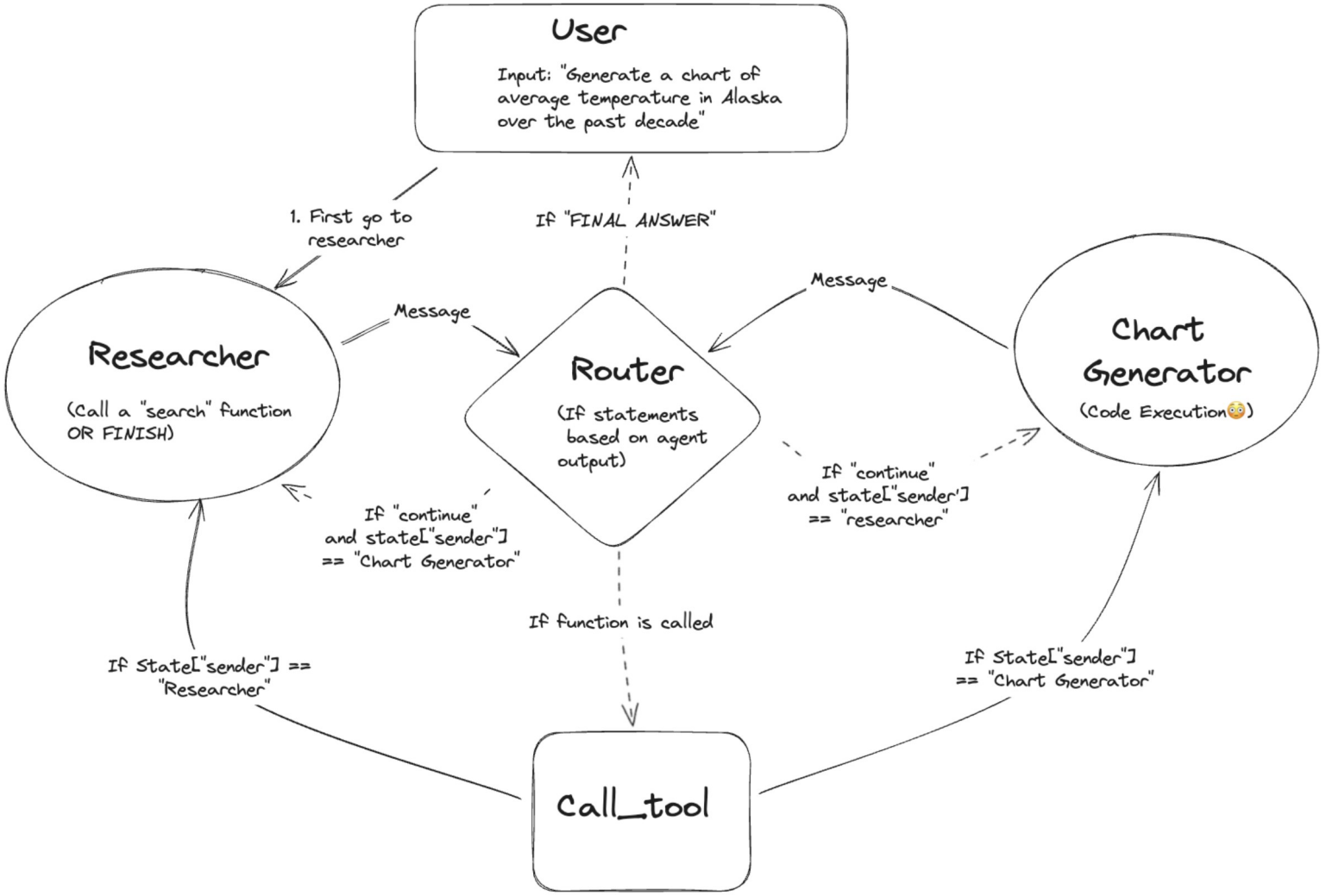
Planning



https://arxiv.org/pdf/2402.02716

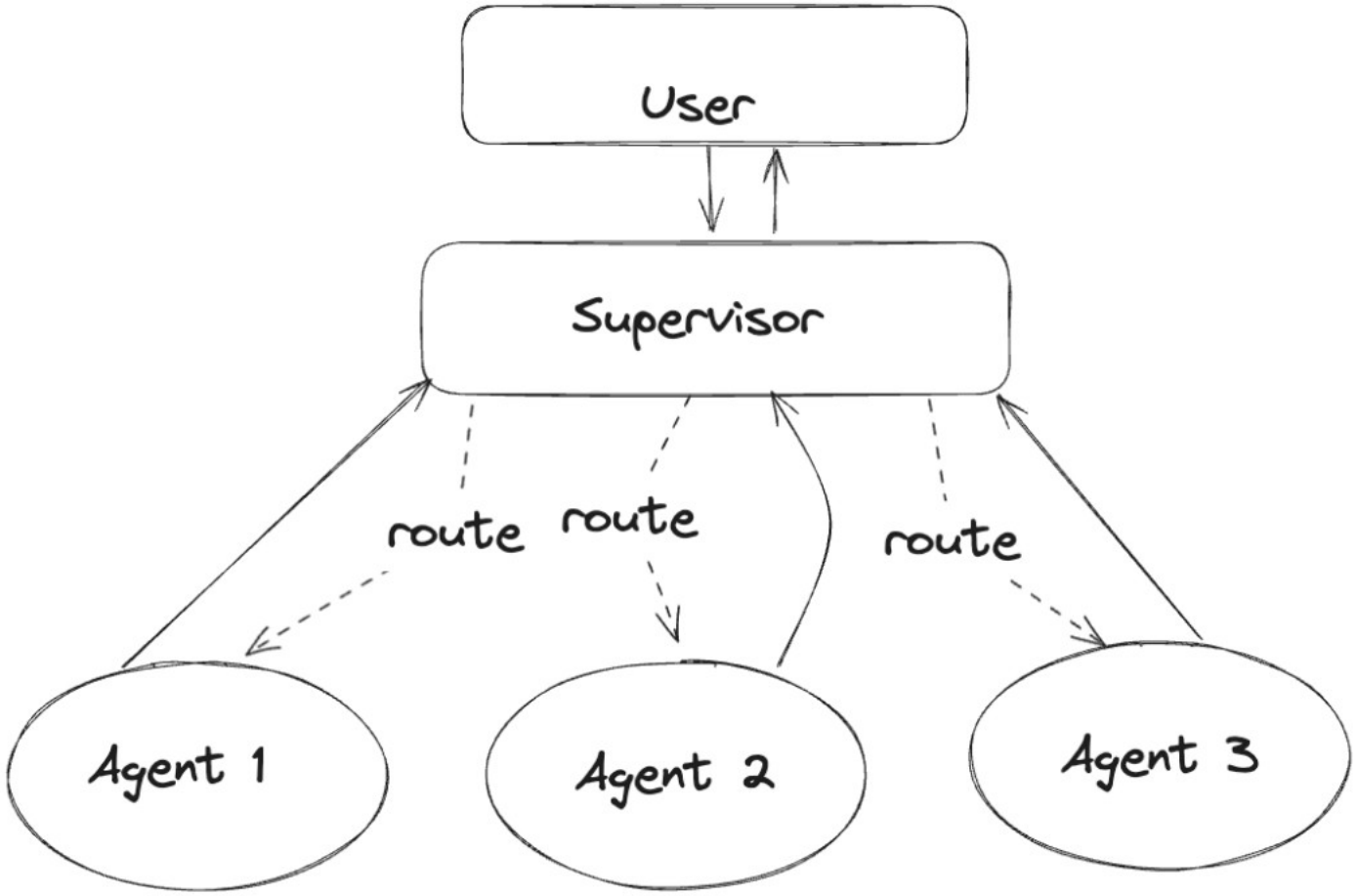
Multi-agent

Collaborative agents



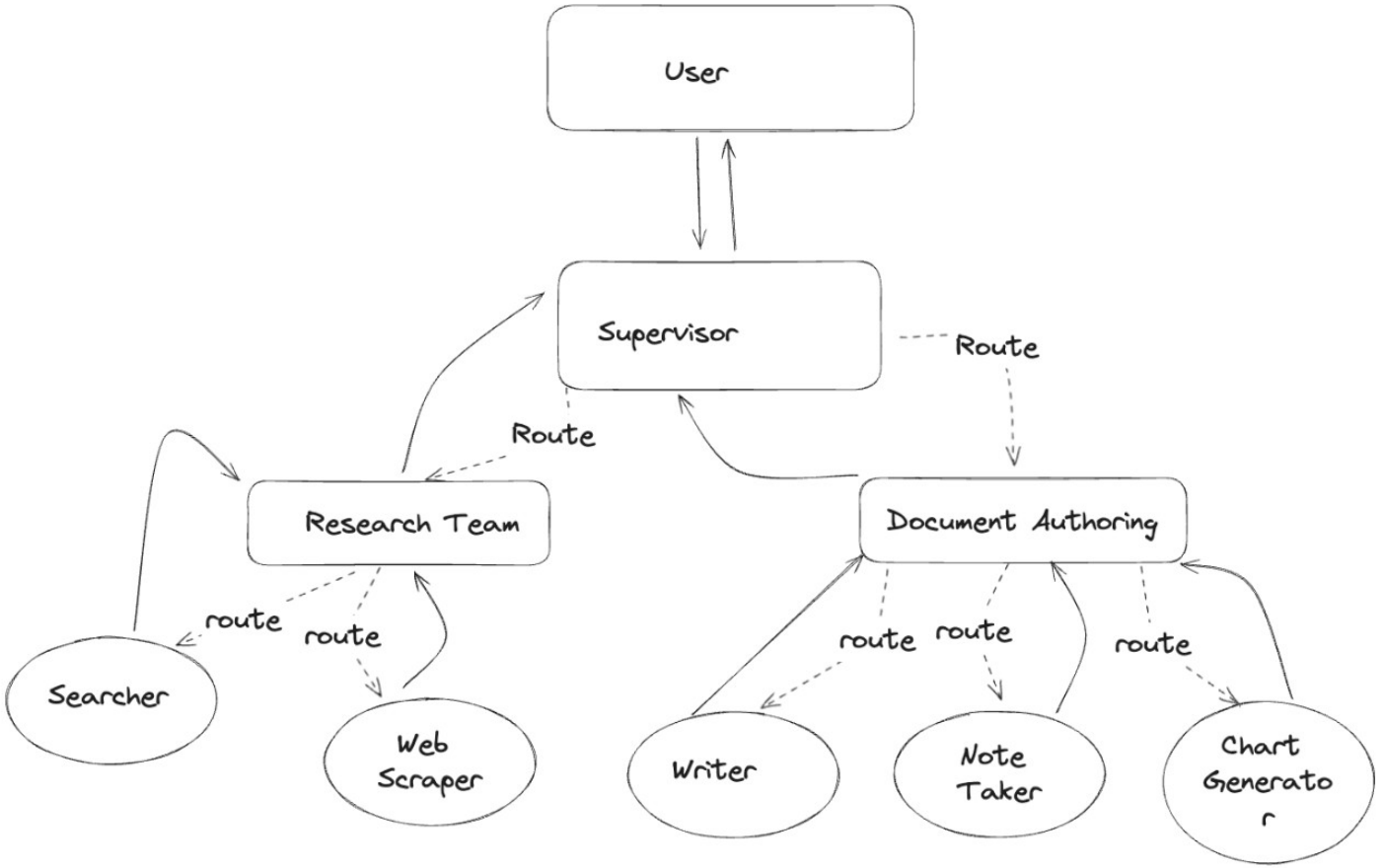
Multi-agent

> With Supervisor Agent



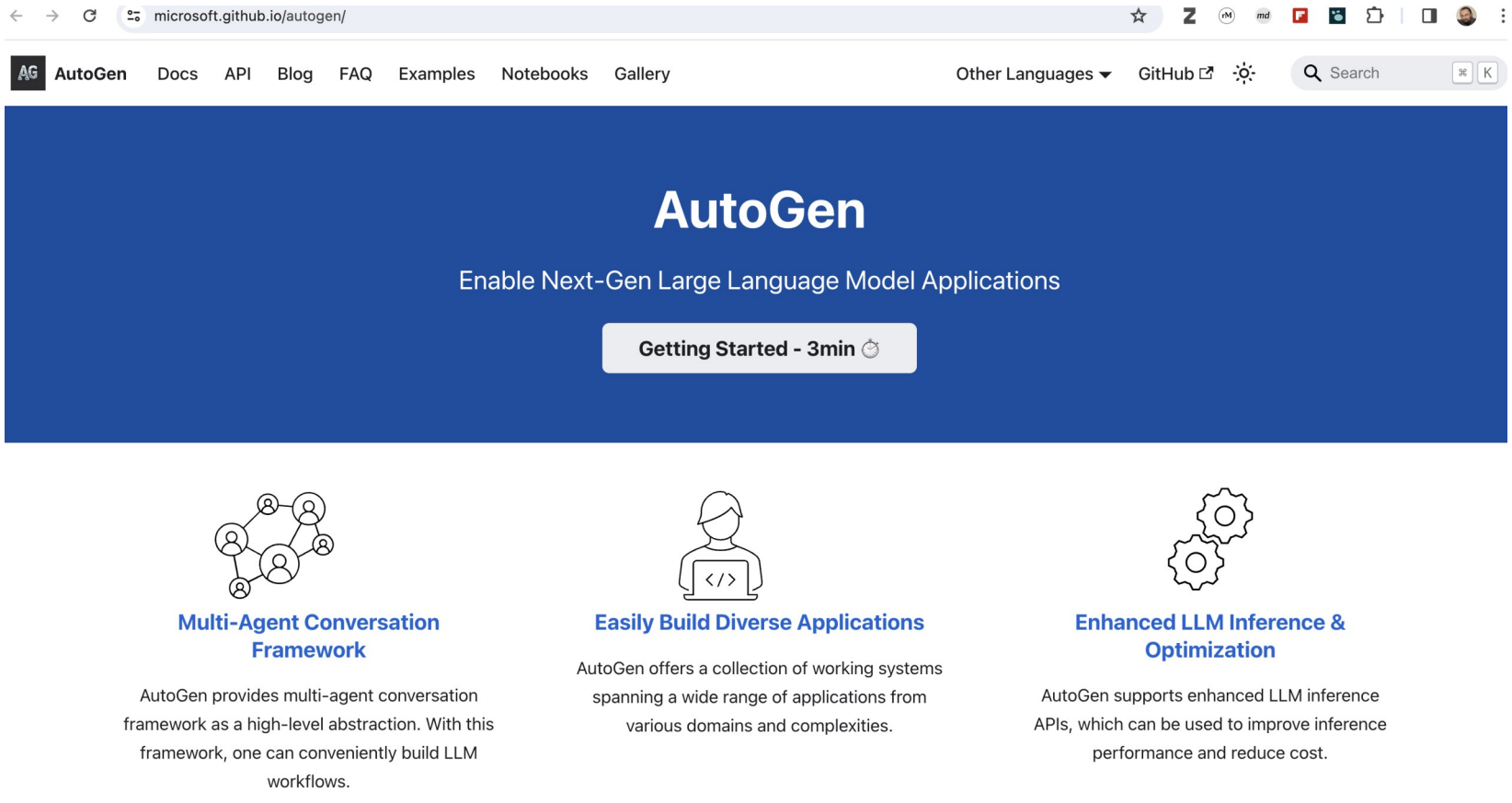
Multi-agent

Hierarchical Teams



Multi-agent

MS Autogen



Multi-agent

> MS Autogen

1. **Code Generation, Execution, and Debugging**
   - Automated Task Solving with Code Generation, Execution & Debugging - View Notebook
   - Automated Code Generation and Question Answering with Retrieval Augmented Agents - View Notebook
   - Automated Code Generation and Question Answering with Qdrant based Retrieval Augmented Agents - View Notebook
2. **Multi-Agent Collaboration (>3 Agents)**
   - Automated Task Solving by Group Chat (with 3 group member agents and 1 manager agent) - View Notebook
   - Automated Data Visualization by Group Chat (with 3 group member agents and 1 manager agent) - View Notebook
   - Automated Complex Task Solving by Group Chat (with 6 group member agents and 1 manager agent) - View Notebook
   - Automated Task Solving with Coding & Planning Agents - View Notebook
   - Automated Task Solving with transition paths specified in a graph - View Notebook
   - Running a group chat as an inner-monolgue via the SocietyOfMindAgent - View Notebook
   - Running a group chat with custom speaker selection function - View Notebook
3. **Sequential Multi-Agent Chats**
   - Solving Multiple Tasks in a Sequence of Chats Initiated by a Single Agent - View Notebook
   - Async-solving Multiple Tasks in a Sequence of Chats Initiated by a Single Agent - View Notebook
   - Solving Multiple Tasks in a Sequence of Chats Initiated by Different Agents - View Notebook
4. **Nested Chats**
   - Solving Complex Tasks with Nested Chats - View Notebook
   - OptiGuide for Solving a Supply Chain Optimization Problem with Nested Chats with a Coding Agent and a Safeguard Agent - View Notebook
5. **Applications**
   - Automated Chess Game Playing & Chitchatting by GPT-4 Agents - View Notebook
   - Automated Continual Learning from New Data - View Notebook
   - OptiGuide - Coding, Tool Using, Safeguarding & Question Answering for Supply Chain Optimization
   - AutoAnny - A Discord bot built using AutoGen
6. **Tool Use**
   - **Web Search**: Solve Tasks Requiring Web Info - View Notebook
   - Use Provided Tools as Functions - View Notebook
   - Use Tools via Sync and Async Function Calling - View Notebook
   - Task Solving with Langchain Provided Tools as Functions - View Notebook
   - **RAG**: Group Chat with Retrieval Augmented Generation (with 5 group member agents and 1 manager agent) - View Notebook
   - Function Inception: Enable AutoGen agents to update/remove functions during conversations. - View Notebook
   - Agent Chat with Whisper - View Notebook
   - Constrained Responses via Guidance - View Notebook
   - Browse the Web with Agents - View Notebook
   - **SQL**: Natural Language Text to SQL Query using the Spider Text-to-SQL Benchmark - View Notebook
7. **Human Involvement**
   - Simple example in ChatGPT style View example
   - Auto Code Generation, Execution, Debugging and **Human Feedback** - View Notebook
   - Automated Task Solving with GPT-4 + **Multiple Human Users** - View Notebook
   - Agent Chat with **Async Human Inputs** - View Notebook
8. **Agent Teaching and Learning**
   - Teach Agents New Skills & Reuse via Automated Chat - View Notebook
   - Teach Agents New Facts, User Preferences and Skills Beyond Coding - View Notebook
   - Teach OpenAI Assistants Through GPTAssistantAgent - View Notebook
   - Agent Optimizer: Train Agents in an Agentic Way - View Notebook
9. **Multi-Agent Chat with OpenAI Assistants in the loop**
   - Hello-World Chat with OpenAi Assistant in AutoGen - View Notebook
   - Chat with OpenAI Assistant using Function Call - View Notebook
   - Chat with OpenAI Assistant with Code Interpreter - View Notebook
   - Chat with OpenAI Assistant with Retrieval Augmentation - View Notebook
   - OpenAI Assistant in a Group Chat - View Notebook
10. **Multimodal Agent**
    - Multimodal Agent Chat with DALLE and GPT-4V - View Notebook
    - Multimodal Agent Chat with Llava - View Notebook
    - Multimodal Agent Chat with GPT-4V - View Notebook
11. **Long Context Handling**
    - Long Context Handling as A Capability - View Notebook
12. **Evaluation and Assessment**
    - AgentEval: A Multi-Agent System for Assess Utility of LLM-powered Applications - View Notebook
13. **Automatic Agent Building**
    - Automatically Build Multi-agent System with AgentBuilder - View Notebook
    - Automatically Build Multi-agent System from Agent Library - View Notebook

Multi-agent

> ChatDev

github.com/OpenBMB/ChatDev

README | Apache-2.0 license

Communicative Agents for Software Development

【English | Chinese | Japanese | Korean | Filipino | French | Slovak | Portuguese | Spanish | Dutch | Turkish | Hindi | Bahasa Indonesia】

【Wiki | Visualizer | Community Built Software | Customization | Discord】

Overview

- **ChatDev** stands as a **virtual software company** that operates through various **intelligent agents** holding different roles, including Chief Executive Officer, Chief Product Officer, Chief Technology Officer, programmer, reviewer, tester, art designer. These agents form a multi-agent organizational structure and are united by a mission to "revolutionize the digital world through programming." The agents within ChatDev **collaborate** by participating in specialized functional seminars, including tasks such as designing, coding, testing, and documenting.
- The primary objective of ChatDev is to offer an **easy-to-use**, **highly customizable** and **extendable** framework, which is based on large language models (LLMs) and serves as an ideal scenario for studying collective intelligence.

Multi-agent

CrewAI

crewai.com

crewai

Documentation | Chat with our docs | Join our Discord | Blog | Login | Sign Up

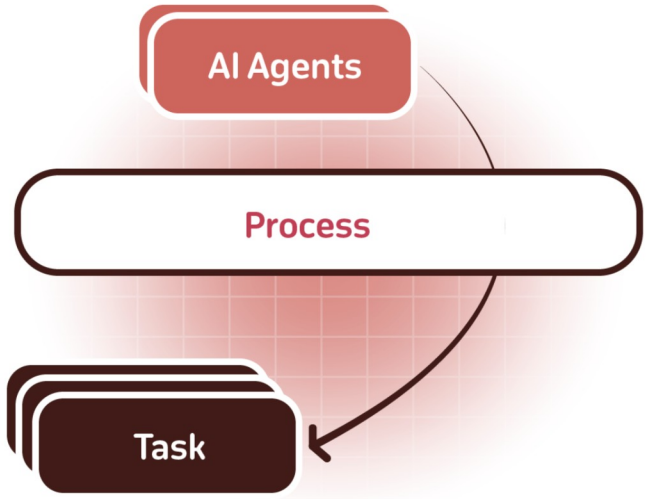
AI Agents for real use cases

Most AI Agents frameworks are complex
We provide the power with simplicity
We provide a platform and hope you create wonders.

1,400,000+
Multi-Agent Crews run last 7 days using crewAI.

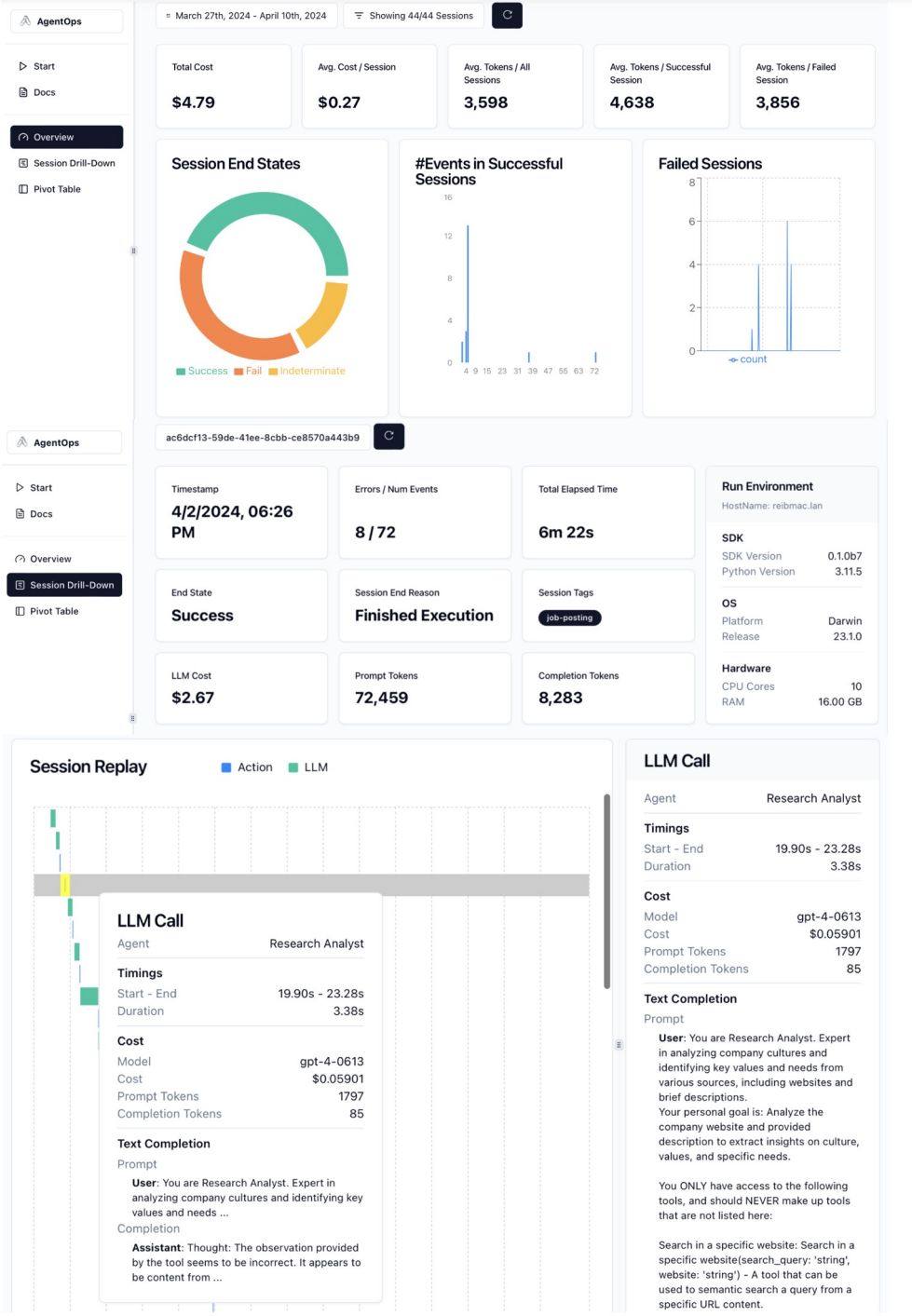
CrewAI+ for Enterprises

The framework | Documentation | Chat with our docs



Multi-Agent Automations for Production

Become a certified crewAI Engineer

Multi-agent

CrewAI